

The Digestive System

A tube that breaks food into pieces small enough to cross a membrane. Nothing counts until it does.

BY THE END OF THIS PAGE YOU SHOULD BE ABLE TO

- List the six jobs the digestive system does
- Say which nutrient is broken down where, and by what
- Explain why the small intestine does nearly all the absorbing
- Explain what bile does, and why it is not an enzyme
- Say why everything absorbed from the gut goes to the liver first

1 Six jobs, one tube

Food inside the digestive tract is still, technically, outside the body.

The tract is a tube running through you, open at both ends. Nothing in it has entered your body until it crosses the lining and reaches the blood.

That one idea explains the whole design. Everything before absorption exists to make absorption possible.

- 1 **Ingestion.** Taking food in.
- 2 **Propulsion.** Moving it along.
- 3 **Mechanical digestion.** Chewing and churning. Breaking things up without changing the chemistry.
- 4 **Chemical digestion.** Enzymes splitting large molecules into small ones.
- 5 **Absorption.** Nutrients crossing into blood or lymph. This is the point of the whole thing.
- 6 **Elimination.** Getting rid of what is left.

Two kinds of movement, and they are not the same. Peristalsis is a travelling wave that *moves* contents along. Segmentation is local squeezing that *mixes* contents with enzymes and pushes them against the wall, without moving them onward. The small intestine spends most of its time on segmentation, because absorption takes time.

2 What happens where

Click through the tract and follow a meal.

FOLLOW A MEAL THROUGH THE TRACT

Mouth

Oesophagus

Stomach

Liver & gallbladder

Pancreas

Small intestine

Large intestine

Mouth

Mechanical digestion by chewing, and the only place carbohydrate digestion begins — salivary amylase starts splitting starch. Saliva also lubricates the bolus so it can be swallowed. Nothing is absorbed here.

NUTRIENT	BREAKDOWN STARTS	FINISHED BY	ABSORBED AS
Carbohydrate	Mouth, by saliva	Pancreatic enzymes and gut wall enzymes	Simple sugars
Protein	Stomach, in acid	Pancreatic enzymes	Amino acids and short chains
Fat	Small intestine	Lipase, after bile breaks the fat up	Fatty acids, into lymph rather than blood

3 Why the small intestine does the absorbing

Surface area, and a lot of it.

Three levels of folding multiply the lining. There are folds in the wall, finger-like **villi** on those folds, and a brush of tiny **microvilli** on every cell of every villus.

The result is an absorbing surface far larger than the organ that holds it.

Inside each villus sits a network of capillaries and one lymph vessel. Sugars and amino acids go into the capillary and travel in blood. Digested fats are too big and too greasy for that route, so they are packaged and enter the lymph vessel instead. They reach the bloodstream only near the heart.

WHY THE NURSE CARES

Diseases that flatten the villi, such as celiac disease, do not block digestion. The enzymes still work fine. The patient becomes malnourished because the surface for absorbing the results is gone.

4 Bile, and the problem with fat

Fat does not dissolve in water, and enzymes can only work at a surface.

One large fat droplet in watery gut contents gives lipase very little surface to attack.

Bile, made by the liver and stored in the gallbladder, breaks that droplet into many tiny ones. That hugely increases the surface, without changing a single chemical bond.

So bile is an **emulsifier**, not an enzyme. It does not digest anything. It makes fat digestible by lipase, which does.

Remove the gallbladder and bile still reaches the intestine from the liver. It just arrives steadily instead of in a well-timed squirt, which is why big fatty meals get harder to handle.

THINK OF IT THIS WAY

Washing a greasy pan. The soap does not eat the grease. It breaks it into droplets small enough for the water to carry off. Bile is doing that job.

5 Everything goes to the liver first

Blood leaving the intestine does not rejoin the general circulation. It is routed through the liver.

That route carries nutrient-rich blood from the gut straight to the liver, which checks it before the rest of the body sees it. The liver stores extra sugar, converts amino acids, processes drugs and toxins, and makes plasma proteins and clotting factors.

This is why a pill can be largely broken down before it ever reaches its target, and why the same drug given by injection can be far stronger at the same dose.

It is also why liver failure causes so many different problems. Nutrients, drug clearance, clotting, and fluid balance all pass through one organ.

The **large intestine** finishes the job with far less drama. It absorbs water and some salts, houses bacteria that make a few vitamins, and forms and stores the leftovers. It is a water-recovery operation, which is why upsetting it causes either dehydration or constipation.

6 Check yourself

Five questions. The wrong answers explain themselves, so they are worth reading.

1. Why is food in the stomach not yet considered inside the body?

- a) It has not been chewed enough
- b) It has not crossed the lining into blood or lymph
- c) The stomach is not really an organ
- d) It is still mixed with saliva

2. Which statement about bile is right?

- a) It is an enzyme that digests fat
- b) It breaks fat into small droplets so lipase has more surface
- c) It neutralizes protein in the stomach
- d) The gallbladder makes it

3. Digested fats enter lymph vessels instead of capillaries. What follows from that?

- a) They reach the blood only after travelling in lymph
- b) They are eliminated instead of absorbed
- c) They reach the liver before any other tissue
- d) They are absorbed in the large intestine

4. A patient with celiac disease has flattened villi and becomes malnourished. Why?

- a) Digestive enzymes are no longer made
- b) The absorbing surface is gone
- c) Food no longer moves through the tract
- d) The stomach cannot make acid

5. Why can a pill be much weaker than the same dose given by injection?

- a) Stomach acid always destroys pills
- b) Blood from the intestine passes through the liver first, which breaks the drug down
- c) Drugs cannot be absorbed from the intestine
- d) Digestive juices dilute the drug

Answer key

1. It has not crossed the lining into blood or lymph

Correct. The tract is a tube through the body, and absorption is the moment of entry.

2. It breaks fat into small droplets so lipase has more surface

Correct. It makes fat digestible without digesting it.

3. They reach the blood only after travelling in lymph

Correct. Lymph carries them and empties into the big veins near the heart.

4. The absorbing surface is gone

Correct. Digestion works fine, but there is nowhere for the results to be absorbed.

5. Blood from the intestine passes through the liver first, which breaks the drug down

Correct. Gut blood goes to the liver first, and the liver processes the drug before the body sees it.

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